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Database-driven model predictive control system for online adaptation of an autonomous excavator to environmental conditions

Tomofumi Okada [a,*], Toru Yamamoto [b], Takayuki Doi [c], Kazushige Koiwai [c], Koji Yamashita [c]

[a] Digital Monozukuri (Manufacturing) Education and Research Center, Hiroshima University, 1-4-1, Kagamiyama, Higashi-Hiroshima, Hiroshima, 739-8527, Japan
[b] Graduate School of Advanced Science and Engineering, Hiroshima University, 1-4-1, Kagamiyama, Higashi-Hiroshima, Hiroshima, 739-8527, Japan
[c] Kobelco Construction Machinery Co., Ltd., 2-2-1, Itsukaichikou, Saeki-ku, Hiroshima, 731-5161, Japan

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ABSTRACT

This paper presents the design of a database-driven model predictive control (DD-MPC) system for the online adaptation of autonomous excavators to environmental conditions. Control systems for autonomous excavators should consider environmental conditions as these affect their performance for a given excavation operation. Moreover, these conditions may change during operation. MPC was performed using an excavator-environment interaction model, which was estimated online using DD-Modeling to represent changes in environmental conditions. The target excavation trajectory was modified by predicting excavation motion using MPC and decision based on the prediction to complete a given excavation operation regardless of the environmental conditions. The proposed system was experimentally verified using a radio-controlled excavator, and it was confirmed that a given operation could be completed by adapting to environmental conditions.

1. Introduction

Research and development (R&D) on automatic and autonomous construction machineries has been ongoing (Eraliev, Lee, Shin, & Lee, 2022; Melenbrink, Werfel, & Menges, 2020). These machineries can improve the safety of construction sites, carry out construction using a small workforce, and significantly improve productivity on sites. Furthermore, automatic and autonomous construction using building information modeling (BIM) have been carried out and studied (Kuenzel, Teizer, Mueller, & Blickle, 2016; Tang, Shelden, Eastman, Bozorgi, & Gao, 2019). In this way, digital transformation in the construction industry has been accelerated, and construction methods and machineries have been advanced. However, environmental situations may vary as a construction progresses. Thus, site situations will possibly differ from the plan in advance. Autonomous construction machineries, which allow machines to decide according to current site conditions, are considered necessary for automatic and autonomous construction. Some researches on construction combining BIM and Internet of Things (IoT) were reviewed in Tang et al. (2019). Therefore, to construct such a decision technology, utilizing data obtainable from construction sites and models that represent the machineries and the sites is considered effective. Such data include current information on the machinery itself, work objects, and equipment around the machinery.

As with other construction machineries, R&D is actively being conducted on the automation of hydraulic excavators (Eraliev et al., 2022; Hirano et al., 2018; Kim, Lee, & Seo, 2020; Liu et al., 2022; Sotiropoulos & Asada, 2022). However, difficulties exist due to the unique characteristics of hydraulic excavators. Each construction site is unique, and the work environment and tasks are diverse. Therefore, to achieve high productivity across a wide variety of construction sites, an autonomous excavator must utilize a decision function that enables the work to be executed according to the site conditions.

Focusing on soil excavation, a common operation of hydraulic excavators, the excavation motion is characteristically affected by soil features. In Hirano et al. (2018), the effect of soil features or characteristics on the operations of autonomous excavators was verified. The study mentioned that the control parameters have to be set according to the soil characteristics to demonstrate high control performance. In addition, the soil characteristics may vary from one site to other. They may change during operation owing to changes in soil layers and weather conditions such as rainfall. Therefore, to realize efficient operation with an autonomous excavator, it is important to capture changes in soil characteristics during excavation and automatically adapt the operation to those characteristics. However, in current autonomous excavators, tuning of the controller is often performed according to

* Corresponding author.
E-mail addresses: okadato@hiroshima-u.ac.jp (T. Okada), yama@hiroshima-u.ac.jp (T. Yamamoto), doi.takayuki@kobelco.com (T. Doi), koiwai.kazushige@kobelco.com (K. Koiwai), yamashita.koji2@kobelco.com (K. Yamashita).

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the specific site where the excavator operates. Tuning is generally performed via trial and error, the controllers are suitable only for specific soil characteristics, and responding to changes in such characteristics may be difficult. For example, if such a controller is used, following the target excavation trajectory may be difficult due to changes in soil characteristics; or the excavation operation may stop during excavation. Therefore, the following two points are necessary for the control system of an autonomous excavator: (i) a design that considers the characteristics of the work object, such as the soil and its changes; (ii) the excavation operation must be suitable for the characteristics of the current work object, executable based on the decision of the machinery. Currently, when various construction site data-collection and model-utilization mechanisms are being built, such as the land, infrastructure, transport, and tourism data platforms (Ministry of Land, Infrastructure, Transport and Tourism, 2022a, 2022b), utilizing data and models in constructing decision functions and control systems for autonomous excavators is considered a good idea. By utilizing various construction site data and models, as well as the construction sites and machinery data obtained in real time during excavation, an autonomous excavator can adapt to the current soil characteristics and execute the desired operation.

Modeling is one of the methods used to consider the characteristics of the work object in the control system. To model the excavation operation of an autonomous excavator, a model representing the mechanical and hydraulic system of an excavator and soil collapse is commonly used (Chen, Huo, Liu, Zhang, & Wang, 2022; Jiang, Liao, Zhu, & Hu, 2021). However, the main purpose of such a detailed model is to perform analysis. Furthermore, as the model is complicated, its application to a control system is difficult. The specifications of the computer used for machine control make it difficult to perform calculations using such a model in real time. In addition, in calculating the soil collapse, soil characteristic parameters are required. As the characteristics may change during an operation, measurements at several times during the operation are required to build a model that represents the current soil characteristics. However, measuring the soil characteristics during operation is not preferable from the viewpoint of work efficiency and productivity. Therefore, a simple and low-dimensional model responsive to changes in soil characteristics is desirable. The model uses data easily obtained during operation, especially in a complex excavation phenomenon. Considering that the excavation operation of an autonomous excavator is affected by soil characteristics, their influence will appear cylinder pressure and the posture of the attachment is likely to be affected. The sensors, often installed as a standard, can be used to detect them. Therefore, the relationship between the excavation force and the excavation distance, calculable from the sensor information, is focused on; and it is considered effective for the control system to model the interaction between the hydraulic excavator attachment and the work object. In addition, using a DD approach (Yamamoto, Takao, & Yamada, 2009), the model can be estimated online to capture the interaction's characteristic changes, using data obtained from an autonomous excavator in real time.

Such an interaction model employed by a controller will be effective to carry out an excavation operation aligned with the work object characteristics. Furthermore, hydraulic excavators have hardware limitations, such as thrust generated from each actuator. Therefore, it is desirable to apply a control system that addresses such constraints to an autonomous excavator. As a control method based on a model that can explicitly deal with constraints, MPC studies (Maciejowski, 2002; Wang, 2009) are known. In this method, the future response of the controlled object is predicted using a model, and the control input is determined based on the prediction. Taking the interaction into account, it is thought that a control system that can calculate the appropriate control input according to changes in soil characteristics.

On the other hand, setting target trajectories is also important for efficient excavation with autonomous excavators (Kim, Seo, Um, & Lee, 2021; Yang, Long, Song, Pan, & Zhang, 2021; Zhao, Hu, Liu, Tian, & Xia, 2022). In automatic construction, excavation work is performed with the target excavation trajectories set by the construction plan in advance. In this plan, the soil parameters measured in advance were used to set the excavation trajectory suitable for the site. However, it is difficult to design a plan that takes into account situations such as changes in soil characteristics during the operation. They cannot be measured in advance. It is also difficult to make a plan that considers the emergence of unexpected obstacles in the ground. If such a situation occurs, the autonomous excavator may not be able to operate as instructed when excavating along the preset target trajectory. For example, the autonomous excavator cannot perform at the desired operation velocity, or the operation may stop halfway owing to an obstacle. However, when the situations described occur, in the case of a hydraulic excavator operated manually, the operator can decide whether to adjust the operation according to the soil characteristics and avoid obstacles. Alternatively, it is conceivable that an operator can predict the above situations and change the excavation method by detecting increases in excavation load and changes in soil characteristics based on visual information, vibrations, and sounds of the excavator. Thus, pauses or decrease in work efficiency can be avoided by decision and operating situationally. Therefore, for an autonomous excavator that can adapt to the environment of construction sites and operate according to instructions, a decision function according to the situation and a correspondence function according to the decision result are required, just like an operator does. For example, it is thought that a decision function is necessary to modify the preset target excavation trajectory according to the situation. This function allows the execution of the operation as instructed.

Online trajectory-planning methods for automobiles and mobile robots using MPC have been researched (Scheffe, Henneken, Kloock, & Alrifaee, 2022; Schoels et al., 2020). The application of such a method to an autonomous excavator, to avoid obstacles during lifting and turning, has also been reported (Narikawa, Ishihara, & Kanazawa, 2022). The methods use MPC to predict the motion of a machine to avoid surrounding obstacles, set the predicted motion as the target trajectory, and calculate the control input. In the excavation, it is considered that the excavation trajectory can be modified online similarly. In previous research, the position of surrounding obstacles was detected by external sensing, and the trajectory was determined based on the sensing information. However, as underground obstacle avoidance during excavation was set as a target, the detection of their positions was difficult. Assuming an obstacle in the direction of the excavation, the soil was pushed by the obstacle and the excavation load increased. As the hydraulic excavator-environment interaction model expresses the relationship between the excavation force and excavation progress, if the interaction model obtained during an excavation expresses the above condition, it is considered that the model can be used to predict the slowdown or stoppage of the operation.

This paper proposes the DD-MPC system for autonomous excavators, which considers the environmental conditions utilizing models and data. The excavation control considering environmental conditions is executed using a model that expresses the interaction between the excavator attachment and the work object. The interaction model is updated to represent current environmental conditions using real-time data. In addition, the target excavation trajectory is modified by prediction and decision using the model. The control system is configured to be applicable to multiple links of the attachment excavation operations. The proposed control system enables excavation operations suitable for environmental conditions and the execution of a given excavation. The effectiveness of the proposed control system was verified experimentally using an RC excavator.

2. Overview of the proposed control system

The proposed control system is shown in Fig. 1. The control system consists of three functions: model estimation of the controlled object, model-based control, and trajectory planner, which enable an autonomous excavator to adapt to environmental conditions and achieve

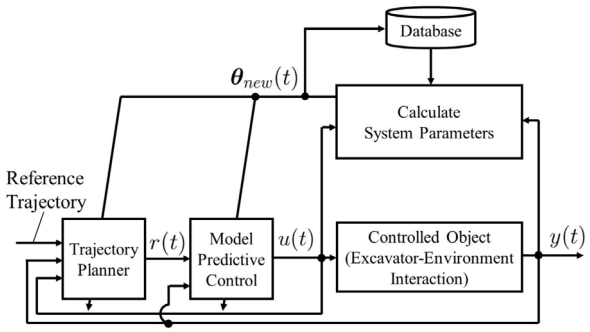


Fig. 1. Block diagram of the database-driven model predictive control system.

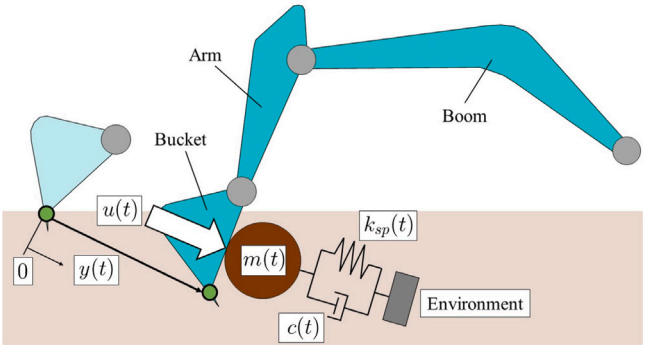


Fig. 2. Hydraulic excavator - environment interaction model.

a given operation. Each of these functions are described as follows. The first function involves model estimation of the controlled object. To consider the work object characteristics of autonomous excavators in the control system, the interaction between the hydraulic excavator attachment and the environment is set as a controlled object; and the interaction is modeled as a low-dimensional one. The control input $u(t)$ and the corresponding output $y(t)$ are defined by the excavation force and the excavation distance, respectively. In addition, to consider the changes in the work object characteristics, the model is estimated using the input/output data of the controlled object and the database with pre-stored models during operation. This method is called DD-Modeling (Okada, Guan, Kinoshita, Yamamoto, Koiwai & Yamashita, 2021). As the timing of updating the model is important, DD-Modeling is executed based on alarm management to update the model (Okada & Yamamoto, 2022). The second function is model based control. The control input is calculated using DD-MPC (Okada, Yamamoto, Doi, Koiwai & Yamashita, 2021; Okada, Yamamoto, Doi, Koiwai, & Yamashita, 2022), which is the MPC (Maciejowski, 2002; Wang, 2009) executed using the models estimated by DD-Modeling. DD-MPC enables the calculation of the control input, which considers the characteristics of the controlled object, while explicitly dealing with its constraints. The control input $u(t)$, calculated using DD-MPC is distributed to the boom and arm (Fig. 2) to move them for excavation by the combined motion of the attachment. The last function is trajectory planning, using the trajectory planner. The trajectory planner modifies the target excavation trajectory in advance to a new trajectory according to the characteristics of the current work object. This modification is performed by predicting the excavation task even when the characteristics of the work object change; for example, when an obstacle appears during the excavation. Modeling of the controlled object, DD-Modeling, the DD-MPC, online excavation trajectory update, and application to an excavation by boom and arm are presented in Sections 3–7.

3. Excavator-environment interaction model

The operation of the hydraulic excavator is affected by the work object characteristics, which may change during operation. Therefore, to adapt to environmental conditions and achieve a given excavation operation autonomously, a control system design considering such changes in work object characteristics is desired. In this section, the controlled object is thus set, and modeled based on the relationship between the input and the output of a controlled object.

To design a control system that considers changes in the characteristics of the work object, a controlled object was set to the interaction between the hydraulic excavator attachment and the environment, and modeled. A hydraulic excavator attachment is often composed of a three-link, consisting of a boom, arm, and bucket, as shown in Fig. 2, and excavation is performed by combining these link operations. In some cases, excavation is performed in combination with the turning motion of the hydraulic excavator body. In this study, excavation by only the attachment operation was considered. The excavator-environment interaction was locally during the excavation operation. Thereby, the relationship between the excavated distance $y(t)$ and the magnitude of the excavation force $u(t)$ was assumed to be a system causing resistance due to the mass, spring, and damper elements. The interaction model is illustrated in Fig. 2, and is expressed by the following equation of motion in continuous-time expression:

$$m(t)\frac{d^2y(t)}{dt^2} + c(t)\frac{dy(t)}{dt} + k_{sp}(t)y(t) = u(t), \tag{1}$$

where $m(t)$, $k_{sp}(t)$, and $c(t)$ denote mass, spring constant, and viscosity coefficient, respectively. Subsequently, the characteristics of the excavator-environment interaction, which change depending on the operating and environmental conditions, are expressed by the changes in the model parameters $m(t)$, $k_{sp}(t)$, and $c(t)$. When excavated with an attachment, it moves in a two-dimensional plane, as shown in Fig. 2. The interaction can also be expressed as a model that considers the direction of each of the two axes; however, in that case, the model becomes complicated, and the number of model parameters increases. Considering that the interaction is locally approximated for simplified and low-dimensional modeling, it is desirable that the model be expressed with as few parameters as possible. Therefore, as shown in the Eq. (1), the motion of the two-dimensional plane is compressed into one-dimensional modeling. The discrete-time system can be described by the following equation, by discretizing Eq. (1) using the finite difference method:

$$\begin{aligned} y(k) = &-a_1(k)y(k-1) - a_2(k)y(k-2) \\ &+ b_0(k)u(k-1) + \xi(k), \end{aligned} \tag{2}$$

where k denotes the step of the discrete-time system, $t = k\,T_s$, and T_s denotes the sampling time. $\xi(k)$ denotes noise and modeling errors. Here, the system parameters $a_1(k)$, $a_2(k)$, and $b_0(k)$ can be obtained by the model parameters $m(k)$, $k_{sp}(k)$, and $c(k)$ by the following equations:

$$a_1(k) = -\frac{2m(k) + c(k)T_s}{m(k) + c(k)T_s + k_{sp}(k)T_s^2}, \tag{3}$$

$$a_2(k) = \frac{m(k)}{m(k) + c(k)T_s + k_{sp}(k)T_s^2}, \tag{4}$$

$$b_0(k) = \frac{T_s^2}{m(k) + c(k)T_s + k_{sp}(k)T_s^2}. \tag{5}$$

4. DD-modeling

This section describes an online modeling method based on database-driven approach. DD-Modeling calculates the model of a controlled object using a database. It has several models stored in advance and the input/output data of the controlled object. In the excavation, the interaction model is estimated online, using the excavation force $u(t)$ and excavation distance $y(t)$ data obtained during control. Here,

considering the model update timing, it is desirable to execute DD-Modeling based on an index from the viewpoint of the control system's design reasonability and calculation cost reduction. Therefore, alarm management (ANSI/ISA-18.2-2016, 2016; Hollifield & Habibi, 2011; Roohi, Chen, Guan, & Yamamoto, 2021) is applied to execute DD-Modeling according to the characteristic change of the control object. Modeling error is used as an evaluation metric in this modeling method. The modeling error is calculated sequentially from the control output $y(t)$ and model output $\hat{y}(t)$. When the error exceeds the threshold, an alarm is issued, and DD-Modeling is executed. As setting the threshold is also important for triggering an alarm, the alarm filter threshold is adaptively adjusted based on the same DD approach.

4.1. Defining query and dataset of DD-modeling

A discrete-time nonlinear system expressed by the following equation was considered:

$$y(k) = h(\boldsymbol{\phi}(k-1)), \tag{6}$$

where $y(k)$ denotes the output, $h(\cdot)$ denotes the nonlinear function, and $\boldsymbol{\phi}(k-1)$ denotes the information vector, respectively. Information vector $\boldsymbol{\phi}(k-1)$ is defined as follows:

$$\boldsymbol{\phi}(k-1) := [y(k-1), \ldots, y(k-n_y), u(k-1), \ldots, u(k-n_u-1)], \tag{7}$$

where $u(k)$ denotes the control input, and n_y and n_u are the orders of output $y(k)$ and input $u(k)$, respectively. The controlled object in Eq. (6) is assumed to be locally described and expressed as follows:

$$A(z^{-1},k)y(k) = B(z^{-1},k)u(k-1) + \xi(k), \tag{8}$$

where $A(z^{-1},k)$ and $B(z^{-1},k)$ are polynomials that describe a discrete-time nonlinear system and are formulated as follows:

$$A(z^{-1},k) = 1 + a_1(k)z^{-1} + \cdots + a_{n_y}(k)z^{-n_y}, \tag{9}$$

$$B(z^{-1},k) = b_0(k) + b_1(k)z^{-1} + \cdots + b_{n_u}(k)z^{-n_u}. \tag{10}$$

When modeling a nonlinear system using local linearization, the model orders n_y and n_u are determined from the relationship between the input and output data obtained in advance from the controlled object. In this study, because the controlled object is locally represented by the interaction model in Eq. (2), $n_y = 2$ and $n_u = 0$. Furthermore, to calculate $A(z^{-1},k)$ and $B(z^{-1},k)$ using Eq. (8) by DD-Modeling, Eqs. (9) and (10) are rewritten as follows:

$$\hat{A}(z^{-1},k) = 1 + \hat{a}_1(k)z^{-1} + \hat{a}_2(k)z^{-2}, \tag{11}$$

$$\hat{B}(z^{-1},k) = \hat{b}_0(k). \tag{12}$$

Using Eqs. (11) and (12), the controlled object in Eq. (8) can be expressed locally as follows:

$$\hat{A}(z^{-1},k)y(k) = \hat{B}(z^{-1},k)u(k-1) + \xi(k). \tag{13}$$

To define the dataset in the database, Eq. (13) was rewritten as:

$$y(k) = \boldsymbol{\theta}(k)\boldsymbol{\psi}(k-1)^{\mathrm{T}} + \xi(k), \tag{14}$$

$$\boldsymbol{\theta}(k) := [-\hat{a}_1(k), -\hat{a}_2(k), \hat{b}_0(k)], \tag{15}$$

$$\boldsymbol{\psi}(k-1) := [y(k-1), y(k-2), u(k-1)]. \tag{16}$$

Here, the system parameters $\boldsymbol{\theta}(k)$ can be described by Eq. (14) as:

$$\boldsymbol{\theta}(k) = f(\tilde{\boldsymbol{\psi}}(k)), \tag{17}$$

$$\tilde{\boldsymbol{\psi}}(k) := [y(k), y(k-1), y(k-2), u(k-1)], \tag{18}$$

where $f(\cdot)$ denotes a linear function and $\tilde{\boldsymbol{\psi}}(k)$ denotes a vector composed of actual measurement data. To calculate the local system parameters $\boldsymbol{\theta}(k)$, the query $\bar{\boldsymbol{\phi}}(k)$ and dataset $\bar{\boldsymbol{\theta}}(j)$ to be stored in the database are defined as follows:

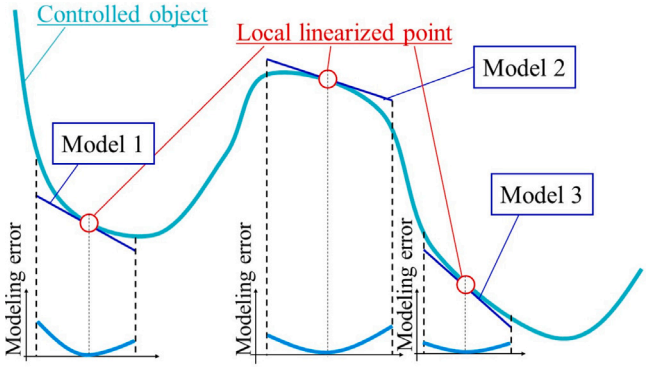


Fig. 3. Modeling errors of each local linear model.

$$\bar{\boldsymbol{\phi}}(k) := [y(k), y(k-1), y(k-2), u(k-1)], \tag{19}$$

$$\bar{\boldsymbol{\theta}}(j) = [-\hat{a}_1(j), -\hat{a}_2(j), \hat{b}_0(j)]. \tag{20}$$

As described above, the system parameters were defined as a dataset, and a database was constructed.

4.2. Alarm filter

Alarm management is known as one of the techniques for responding to abnormalities in industrial processes (ANSI/ISA-18.2-2016, 2016; Hollifield & Habibi, 2011; Roohi et al., 2021). This is a method of designing alarms to detect process abnormalities and respond appropriately and promptly to ensure safe operation. In a control system that utilizes data and models, an increase in modeling error leads to deterioration of the control performance. When applying alarm management to the control system, an increase in the modeling error can be regarded as an alarm; and modeling execution to update an appropriate model can be regarded as a response to the alarm. Applying alarm management to DD-Modeling allows modeling to be performed as needed.

An alarm filter was designed in this study. To enable control for good performance, the modeling error was used as an evaluation index, and an alarm filter was used to detect an increase in the modeling error. The alarm filter is defined as follows:

$$a_e(k) = \begin{cases} 1 & (e(k) > t_{h_{new}}(k)) \\ 0 & (\text{otherwise}) \end{cases}, \tag{21}$$

$$e(k) = |y(k) - \hat{y}(k)|, \tag{22}$$

where $e(k)$, $\hat{y}(k)$, and $t_{h_{new}}(k)$ denote the modeling error, model output, and a threshold of the modeling error, respectively. As the sign of the modeling error is irrelevant, $e(k)$ is defined as the absolute value of the difference between the system output $y(k)$ and model output $\hat{y}(k)$. Subsequently, an alarm was triggered when $a_e(k) = 1$ in Eq. (21), and DD-Modeling was executed.

In designing an alarm filter, setting the threshold $t_{h_{new}}(k)$ used for alarm detection is important. In this study, the excavator attachment-environment interaction, which has nonlinear characteristics, was set as a controlled object, and the interaction was assumed to be locally linearized around each equilibrium point. In addition, the interaction was modeled using Eq. (2). For example, as shown in Fig. 3, the modeling error around each equilibrium point of a nonlinear system is different. Therefore, it was necessary to set a suitable threshold for each model. Using the mean μ_{th} and standard deviation σ_{th} of the modeling error, the threshold t_{hm} for each model is defined by the three-sigma rule. It indicates that in most cases, the modeling error is within the threshold t_{hm}; i.e., if the operating point is around the equilibrium point in the model expressing the equilibrium point. μ_{th} and σ_{th} are calculated based on the modeling error $e(k)$ which is calculated using the data $y(k)$ used for system identification and $\hat{y}(k)$. The latter is calculated using the identified system parameters.

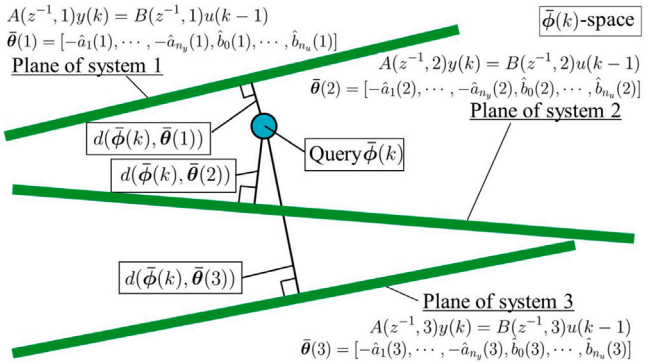


Fig. 4. Distance between query and datasets' plane in $\bar{\boldsymbol{\phi}}(k)$-space.

4.3. Calculation of system parameters and adaptive threshold

The calculation rules for the system parameters and adaptive threshold based on the database-driven approach are described. This rule consists of the following three steps:

[Step 1]*Generating the initial database*

To construct the initial database, the system parameters $\theta(j)$ are obtained by system identification using the input/output data of the controlled object offline. In addition, each model threshold $t_{hm}(j)$ is calculated from the error $e(t)$ between the output data and model output calculated using $\theta(j)$. Subsequently, $\theta(j)$ and $t_{hm}(j)$ are stored in the initial database $\bar{\boldsymbol{\Theta}} \in \mathbb{R}^{N_0\times 4}$, which is defined as follows:

$$\bar{\boldsymbol{\Theta}} = \begin{bmatrix} \bar{\boldsymbol{\theta}}(1) & t_{hm}(1) \\ \vdots & \vdots \\ \bar{\boldsymbol{\theta}}(N_0) & t_{hm}(N_0) \end{bmatrix}, \tag{23}$$

where N_0 denotes the number of datasets in the initial database.

[Step 2]*Calculating the system parameters and adaptive threshold*

In the $\bar{\boldsymbol{\phi}}(k)$-space, the distance between query $\bar{\boldsymbol{\phi}}(k)$ and each dataset $\bar{\boldsymbol{\theta}}(j)$ in the database is as follows:

$$d(\bar{\boldsymbol{\phi}}(k),\bar{\boldsymbol{\theta}}(j)) = \frac{|\sum_{i=1}^{3}\bar{\theta}_i(j)\bar{\phi}_{i+1}(k) - y(k)|}{\sqrt{\sum_{i=1}^{3}\bar{\theta}_i(j)^2 + 1}}, \quad j = 1,\ 2,\ \ldots,\ N(k) \tag{24}$$

where $N(k)$ is the number of datasets stored in the database when $\bar{\boldsymbol{\phi}}(k)$ is obtained, and i denotes the ith element of the query and the dataset. Eq. (24) expresses the distance between query $\bar{\boldsymbol{\phi}}(k)$ and the hyperplane determined by dataset $\bar{\boldsymbol{\theta}}(j)$ and Eq. (8), as shown in Fig. 4. For example, when $d(\bar{\boldsymbol{\phi}}(k),\ \bar{\boldsymbol{\theta}}(j)) = 0$, the system parameters that represent the current-controlled object match $\bar{\boldsymbol{\theta}}(j)$ in the database. Then, the dataset is sorted in ascending order of distance using Eq. (24), and k_n datasets were selected in the order of decreasing distance. The weight w_i of each k_n-piece neighbor dataset is as follows:

$$w_j = \frac{d(\bar{\boldsymbol{\phi}}(k),\ \bar{\boldsymbol{\theta}}(j))^{-n_w}}{\sum_{l=1}^{k_n} d(\bar{\boldsymbol{\phi}}(k),\ \bar{\boldsymbol{\theta}}(l))^{-n_w}}, \quad \sum_{i=1}^{k_n} w_i = 1, \quad j = 1,\ 2,\ \ldots,\ k_n, \tag{25}$$

where n_w denotes the user-specified parameter that distinguishes the difference in weight according to distance. The system parameters and threshold can be obtained with a linear weighted average (Atkeson, Moore, & Schaal, 1997) by using the following equations:

$$\boldsymbol{\theta}_{new}(k) = \sum_{i=1}^{k_n} w_i\bar{\boldsymbol{\theta}}(i), \tag{26}$$

$$t_{h_{new}}(k) = \sum_{i=1}^{k_n} w_i t_{hm}(i). \tag{27}$$

[Step 3]*Removing redundant data*

The storage capacity consumption and calculation cost of the computer increase as the number of datasets $N(k)$ in the database increase. Therefore, to reduce datasets, it is desirable to remove redundant data from the database. Thus, a model is very similar to that newly calculated using Eq. (26). Of all datasets, the ones that satisfied the following equation were selected and deleted.

$$d_j(\bar{\boldsymbol{\theta}}(j),\boldsymbol{\theta}_{new}(k)) = \sum_{i=1}^{3}|\bar{\theta}_i(j) - \theta_{new,i}(k)| \le \beta, \tag{28}$$

where β denotes a user-specified parameter for selecting datasets to be deleted. Eq. (28) is purposed to select datasets in the database highly similar to the system parameters calculated by Eq. (26). Here, if multiple datasets correspond to Eq. (28), the dataset with the smallest distance is deleted.

By performing [Step 2] and [Step 3] when $a_e(k) = 1$, the system parameters $\boldsymbol{\theta}_{new}(k)$ representing the current controlled object and threshold $t_{h_{new}}(t)$ are calculated online. They are applied to the model in the controller in Fig. 1 and the adaptive threshold of the alarm filter.

5. DD-MPC

This section describes the construction of DD-MPC by combining MPC and DD-Modeling to handle controlled objects whose characteristics change. Control input $u(k)$ is defined as follows:

$$u(k) = u(k-1) + \Delta u(k), \tag{29}$$

where $\Delta u(k)$ denotes the input variation, and is calculated by MPC. To obtain the desired control performance even for controlled objects whose characteristics change during operation, the MPC responds to changes in the characteristics of the controlled object using system parameters estimated by DD-Modeling. To apply the excavator-environment interaction model to DD-MPC, Eq. (2) is replaced with the estimated system parameters $\hat{a}_1(k)$, $\hat{a}_2(k)$, and $\hat{b}_0(k)$, and represented by a state-space model as follows:

$$\boldsymbol{x}(k+1) = \boldsymbol{A}_s(k)\boldsymbol{x}(k) + \boldsymbol{B}_s(k)\Delta u(k), \tag{30}$$

$$y(k) = \boldsymbol{C}_s\boldsymbol{x}(k), \tag{31}$$

$$\boldsymbol{x}(k) := [\Delta y(k),\ \Delta y(k-1),\ y(k)]^{\mathrm{T}}, \tag{32}$$

$$\boldsymbol{A}_s(k) = \begin{bmatrix} -\hat{a}_1(k) & -\hat{a}_2(k) & 0 \\ 1 & 0 & 0 \\ -\hat{a}_1(k) & -\hat{a}_2(k) & 1 \end{bmatrix}, \tag{33}$$

$$\boldsymbol{B}_s(k) = [\hat{b}_0(k),\ 0,\ \hat{b}_0(k)]^{\mathrm{T}}, \tag{34}$$

$$\boldsymbol{C}_s = [0,\ 0,\ 1], \tag{35}$$

where $\boldsymbol{A}_s(k)$, $\boldsymbol{B}_s(k)$, and $\boldsymbol{C}_s$ denote the system, control, and observation matrices, respectively. $\hat{a}_1(k)$, $\hat{a}_2(k)$, and $\hat{b}_0(k)$ calculated by DD-Modeling are applied to $\boldsymbol{A}_s(k)$ and $\boldsymbol{B}_s(k)$. Then, to obtain $\Delta u(k)$, the evaluation function J is defined by the following equation using the state-space model of Eqs. (30), (31):

$$J = (\boldsymbol{y}_p - \boldsymbol{r}_p)^{\mathrm{T}}\boldsymbol{Q}_e(\boldsymbol{y}_p - \boldsymbol{r}_p) + \Delta\boldsymbol{u}_p^{\mathrm{T}}\boldsymbol{Q}_u\Delta\boldsymbol{u}_p, \tag{36}$$

$$\boldsymbol{y}_p = [y(k+1),\ldots,\ y(k+N_p)]^{\mathrm{T}}, \tag{37}$$

$$\boldsymbol{r}_p = [r(k+1),\ldots,\ r(k+N_p)]^{\mathrm{T}}, \tag{38}$$

$$\Delta\boldsymbol{u}_p = [\Delta u(k),\ldots,\ \Delta u(k+Nc-1)]^{\mathrm{T}}, \tag{39}$$

$$\boldsymbol{Q}_e = w_e\boldsymbol{I}_e \in \mathbb{R}^{N_p\times N_p}, \tag{40}$$

$$\boldsymbol{Q}_u = w_u\boldsymbol{I}_e \in \mathbb{R}^{N_c\times N_c}, \tag{41}$$

where $\boldsymbol{y}_p$ is the predicted output, $\boldsymbol{r}_p$ is the target value, $\boldsymbol{u}_p$ is the predicted input; N_p and N_c are the predicted horizon and control horizon, and $\boldsymbol{Q}_e$ and $\boldsymbol{Q}_u$ are weight matrices, respectively. Here, the control performance changes depending on the settings N_p, N_c, $\boldsymbol{Q}_e$, and $\boldsymbol{Q}_u$. For example, the rise to the target value will be gradual if N_p or $\boldsymbol{Q}_u$

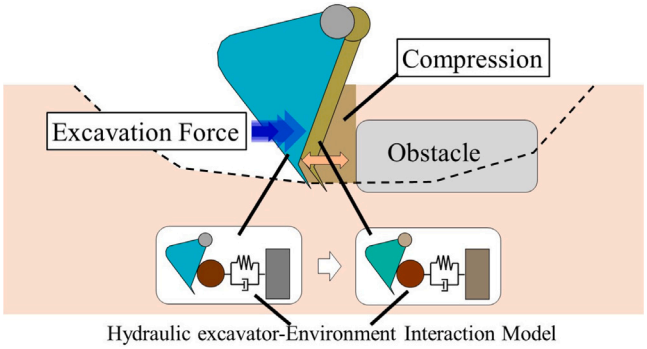


Fig. 5. Prediction in hydraulic excavator-environment interaction.

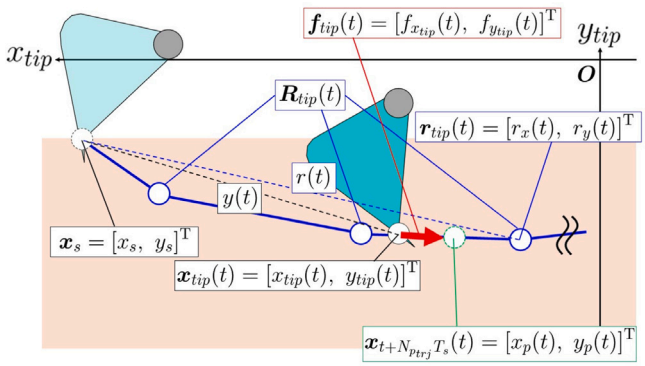


Fig. 6. Geometric relationship of excavation in the proposed control system.

are set to large values. $\boldsymbol{y}_p$ can be obtained recursively using the state space models (30) and (31). $N_c \leq N_p$, $\boldsymbol{u}_p$ is assumed to be constant from N_c to N_p steps ahead. Subsequently, the following constrained quadratic programming problem is solved to calculate $\Delta \boldsymbol{u}_p$:

$$\min_{\Delta \boldsymbol{u}_p} \ J, \tag{42}$$

subject to

$$u_{min} \leq u(k) \leq u_{max}, \tag{43}$$

$$\Delta u_{min} \leq \Delta u(k) \leq \Delta u_{max}, \tag{44}$$

where u_{max} and u_{min} denote the maximum and minimum values of $u(k)$, respectively. Δu_{max} and Δu_{min} are the maximum and minimum values of $\Delta u(k)$, respectively. Finally, $\Delta u(k)$ is obtained using the first element of $\Delta \boldsymbol{u}_p$ as follows:

$$\Delta u(k) = [1,\ 0, \ldots,\ 0]\Delta \boldsymbol{u}_p. \tag{45}$$

As described above, the control input $u(k)$ is obtained using Eq. (29) and (45), respectively.

6. Online excavation trajectory update

Based on DD-MPC, an online updating method for the excavation trajectory is described in this section. The details of the online excavation-trajectory update method based on the DD-MPC are described below. First, the excavation motion is predicted by the DD-MPC using a model that considers the soil characteristics (see Fig. 5). Next, based on the motion prediction, the stop decision of the motion is conducted; and the target trajectory is modified by setting the approximate obstacle position based on the decision result.

This control system considers excavation operation in the two-dimensional plane shown in Fig. 6. Here, the control output $y(t)$ denotes the one-dimensional values of the distance from the excavation start point to the bucket tip position. Thus, $y(t)$ is calculated by the following equation using the bucket tip position $\boldsymbol{x}_{tip}(t) = [x_{tip}(t),\ y_{tip}(t)]^{\mathrm{T}}$ and excavation start point $\boldsymbol{x}_s = [x_s,\ y_s]^{\mathrm{T}}$:

$$y(t) = \sqrt{(x_{tip}(t) - x_s)^2 + (y_{tip}(t) - y_s)^2}. \tag{46}$$

The target trajectory is modified using the trajectory planner shown in Fig. 1, and the target value $r(t)$ is applied to the DD-MPC. Then, $r(t)$, which denotes the distance from the excavation start point to the target position, is defined as follows, using $\boldsymbol{r}_{tip}(t) = [r_x(t),\ r_y(t)]^{\mathrm{T}}$ in the target trajectory $\boldsymbol{R}_{tip}(t)$ and $\boldsymbol{x}_s$:

$$r(t) = \sqrt{(r_x(t) - x_s)^2 + (r_y(t) - y_s)^2}. \tag{47}$$

$\boldsymbol{R}_{tip}(t)$ is defined as a matrix with N_{tip} target positions, and the target position $r_{tip}(t)$ is defined as follows:

$$\boldsymbol{r}_{tip}(t) = \boldsymbol{R}_{tip,j_t}(t) \in \mathbb{R}^{N_{tip} \times 2} \tag{48}$$
$$j_t = 1,\ 2,\ \ldots,\ N_{tip}.$$

In the control system shown in Fig. 1, when the bucket tip position $\boldsymbol{x}_{tip}(t)$ reaches near the current target position $\boldsymbol{R}_{tip,j_t}(t)$, target position $\boldsymbol{r}_{tip}(t)$ is updated to the next target position $\boldsymbol{R}_{tip,j_t+1}(t)$.

The online excavation-trajectory update method consists of "prediction and decision [Step 1t]" and "modification [Step 2t]".

[Step 1t] *Movement prediction of bucket tip position*

$N_{p_{trj}}$ steps ahead, control output $y(t + N_{p_{trj}}T_s)$ is calculated by the DD-MPC using the hydraulic excavator-environment interaction model of Eq. (2). Considering that the equation is a locally linearized model of the interaction, and the interaction changes during operation, the model is used to predict the most recent movement of the bucket tip $\boldsymbol{x}_{tip}(t)$ during excavation. Therefore, movement prediction is performed up to the next target position $\boldsymbol{r}_{tip}(t)$. Furthermore, as the prediction was performed using the one-dimensional simplified model of Eq. (2), complex movement prediction on a two-dimensional plane in the longitudinal direction of the attachment cannot be performed. Therefore, the bucket tip position movement is assumed to move linearly from the current position $\boldsymbol{x}_{tip}(t)$ to the next target position $\boldsymbol{r}_{tip}(t)$ during the operation, as shown in Fig. 6; and prediction using the interaction model is performed. Provided that output $y(t)$ of the model in Eq. (2) is the excavation distance from the excavation starting point, that is, the one-dimensional value. In predicting the movement to the target position $\boldsymbol{r}_{tip}(t)$, the bucket tip position $\boldsymbol{x}_{t+N_{p_{trj}}T_s}(t)$ on the two-dimensional plane is calculated using the predicted output $y(t + N_{p_{trj}}T_s)$ and the excavation start point $\boldsymbol{x}_s$ according to the geometrical relationship shown in Fig. 6. Subsequently, in movement prediction during $N_{p_{trj}}$ steps, it is determined whether the bucket tip position can reach the next target position. If it is unreachable, the target trajectory is modified in [Step 2t].

[Step 2t] *Modification of the target excavation trajectory*

If the bucket tip position $\boldsymbol{x}_{tip}(t)$ cannot reach the next target position $\boldsymbol{r}_{tip}(t)$ in the prediction interval of [Step 1t], the target trajectory is modified based on the positional relationship between the current bucket-tip position and the target position. An example of each position relation assumed during excavation is shown in Fig. 7. Under the condition, it is assumed that an excavation-preventing obstacle appears in the direction of the bucket tip movement, and the target trajectory is modified to avoid the obstacle. Even if no obstacle appears, or the load during excavation increases significantly, the target trajectory is modified by the same method. Encountering hard soil is one such example. The modification method for the example shown in Fig. 7 is described below.

Considering the latter half of the excavation, i.e., when the excavation progresses in the surface direction of the ground, impeding obstacles appear. In this case, to avoid the obstacles and proceed with excavation, the target position is shifted in the x_{tip} positive direction

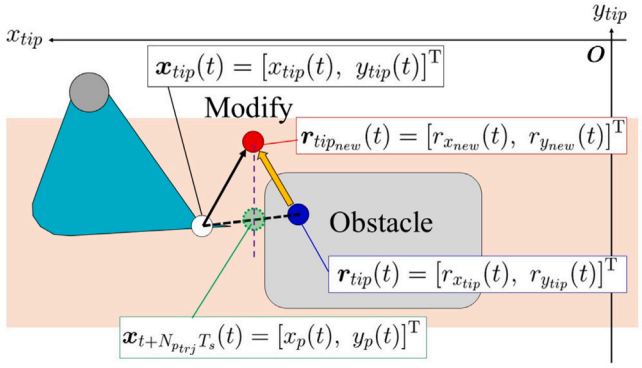


Fig. 7. Geometric relationship between predicted bucket tip position and obstacle.

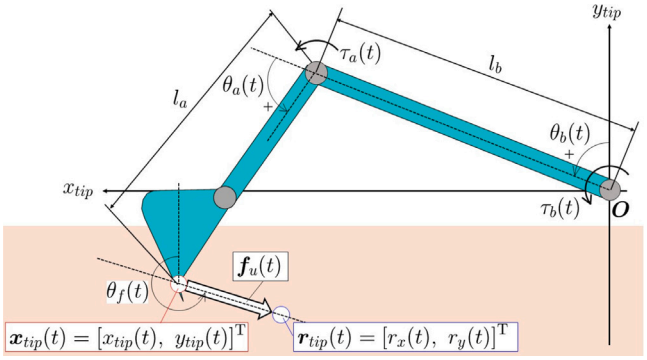


Fig. 8. Excavation with boom and arm.

and y_{tip} upward direction. Here, the modified target position is determined by setting the excavation to proceed to the predicted stop position $y_p(t)$, and also by the distance between the current bucket tip position and target position. This distance is expressed as follows:

$$d_{tip}(t) = \sqrt{(x_{tip}(t) - r_x(t))^2 + (y_{tip}(t) - r_y(t))^2}. \tag{49}$$

The target position is modified by the above assumptions as follows:

$$\boldsymbol{r}_{tip_{new}}(t) = [r_{x_{new}}(t),\ r_{y_{new}}(t)]^{\mathrm{T}}, \tag{50}$$

$$r_{x_{new}}(t) = x_{tip}(t) + \sqrt{d_{tip}(t)^2 - (r_x(t) - y_p(t))^2}, \tag{51}$$

$$r_{y_{new}}(t) = y_p(t). \tag{52}$$

Thereby, modifying the target trajectory allows excavation to continue, avoiding obstacles toward the ground surface and reach the endpoint of the excavation trajectory.

7. Application to excavations by booms and arms

The excavation force $u(t)$ at the bucket tip was calculated using the DD-MPC. However, when excavating with combined of boom and arm movements, command values for each of them are required. As shown in Fig. 8, the bucket tip position $\boldsymbol{x}_{tip}(t)$ advances to the target position $\boldsymbol{r}_{tip}(t)$, and the attachment must generate the excavation force vector in the direction of $\boldsymbol{r}_{tip}(t)$. Therefore, the direction of the excavation force $\theta_f(t)$ was calculated from the geometric relationship of each position; and the excavation force vector $\boldsymbol{f}_u(t)$ was calculated using $u(t)$ and $\theta_f(t)$ as follows:

$$\boldsymbol{f}_u(t) = u(t)[\sin\theta_f(t),\ \cos\theta_f(t)]^{\mathrm{T}}. \tag{53}$$

Subsequently, the torques of the boom and arm required to generate the excavation force vector $\boldsymbol{f}_u(t)$ are calculated and used as their respective command values. The kinematics of the attachment shown in Fig. 8 can be expressed by the follows:

$$\boldsymbol{x}_{tip}(t) = \begin{bmatrix} l_b \sin\theta_b(t) & l_a \sin(\theta_b(t)+\theta_a(t)) \\ l_b \cos\theta_b(t) & l_a \cos(\theta_b(t)+\theta_a(t)) \end{bmatrix} \boldsymbol{q}(t), \tag{54}$$

$$\boldsymbol{q}(t) = [\theta_b(t),\ \theta_a(t)]^{\mathrm{T}}, \tag{55}$$

where l_b denotes boom length and l_a denotes the total length of the arm and bucket; and $\theta_b(t)$ and $\theta_a(t)$ represent the boom and arm angles, respectively. Differentiating Eq. (54) yields the following:

$$\dot{\boldsymbol{x}}_{tip}(t) = \boldsymbol{J}(t)\dot{\boldsymbol{q}}(t), \tag{56}$$

$$\boldsymbol{J}(t) = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}, \tag{57}$$

$$J_{11} = l_b \cos\theta_b(t) + l_a \cos(\theta_b(t) + \theta_a(t)), \tag{58}$$

$$J_{12} = l_a \cos(\theta_b(t) + \theta_a(t)), \tag{59}$$

$$J_{21} = -l_b \sin\theta_b(t) - l_a \sin(\theta_b(t) + \theta_a(t)), \tag{60}$$

$$J_{22} = -l_a \sin(\theta_b(t) + \theta_a(t)). \tag{61}$$

Furthermore, using $\boldsymbol{f}_u(t)$ and $\boldsymbol{J}(t)$, the joint torques of boom and arm are calculated as follows:

$$\boldsymbol{\tau}(t) = \boldsymbol{J}(t)^{\mathrm{T}} \boldsymbol{f}_u(t), \tag{62}$$

$$\boldsymbol{\tau}(t) = [\tau_b(t),\ \tau_a(t)]^{\mathrm{T}}, \tag{63}$$

where $\tau_b(t)$ and $\tau_a(t)$ represent the torques around the axis of rotation of the boom and that of arm, respectively. By moving the boom and arm with $\tau_b(t)$ and $\tau_a(t)$ calculated above, excavation operation is realized in which the bucket tip follows the target trajectory.

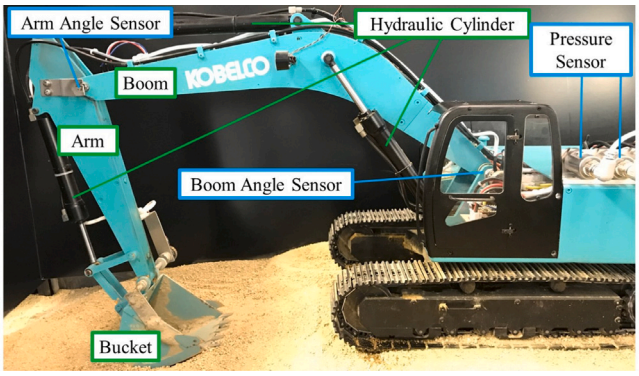


Fig. 9. Radio-controlled excavator.

8. Verification

The proposed control system was applied to an RC excavator, as shown in Fig. 9. It is the 1/12 SCALE EARTH DIGGER 4200XL HYDRAULIC EXCAVATOR manufactured by RC4WD, and was modified by installing sensors to acquire the attachment angles and hydraulic cylinder pressure. The attachment consists of a boom, an arm, and a bucket similar to hydraulic excavators, and each link is operated using hydraulic cylinders. Oil was supplied to each hydraulic cylinder from a pump, and the amount was adjusted by the valve opening. Verification was performed for the excavation operation using a boom and an arm. It was confirmed that the RC excavator could excavate by aligning the bucket tip along the target excavation trajectory; and that the RC excavator could reach the end of the target excavation trajectory even when an obstacle appeared during excavation. In addition, movement execution, as instructed, was defined to reach the endpoint of the target excavation trajectory. In the verification, the control systems with and without the Trajectory Planner (method 1) shown in Fig. 1 were compared.

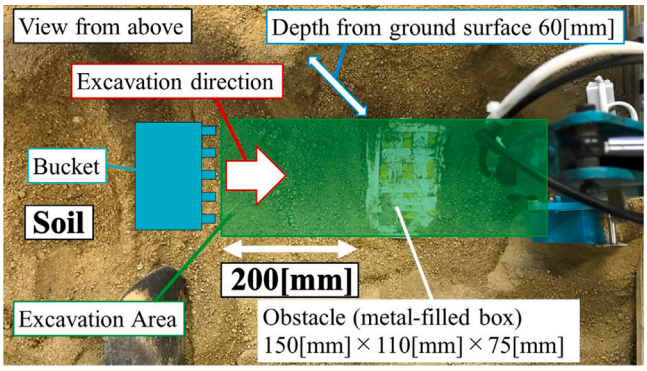


Fig. 10. Experimental condition with buried obstacle.

Table 1
Parameters of the experiments.

Sampling time	$T_s = 0.05$ [s]
System order	$n_y = 2,\ n_u = 0$
MPC horizon	$N_p = 30$ [step], $N_c = 4$ [step]
Weight of cost function	$\boldsymbol{Q}_e = 1\times\boldsymbol{I}_e$, $\boldsymbol{Q}_u = 10\times\boldsymbol{I}_u$
Constraint parameters	$\Delta u_{max} = 10$ [N], $\Delta u_{min} = -10$ [N] $u_{max} = 180$ [N], $u_{min} = 0$ [kN]
Prediction step of trajectory planner	$N_{p_{trj}} = 14$ [step]
Target excavation trajectory ($x_{tip} - y_{tip}$ Coordinates)	$[r_x(t)$ [m], $r_y(t)$ [m]] = [0.78, −0.12], [0.72, −0.14], [0.66, −0.15] [0.60, −0.16], [0.54, −0.15], [0.48, −0.14], [0.42, −0.13], [0.36, −0.10]

The target excavation trajectory and setting parameters used in the experiment are listed in Table 1. The constraint conditions for $u(t)$ and $\Delta u(t)$ were determined based on a prior experiment using the RC excavator and the movement direction of the attachment during verification. With $\boldsymbol{Q}_e$ and $\boldsymbol{Q}_u$ to simplify the setting, $\boldsymbol{Q}_e$ set as a unit matrix and $\boldsymbol{Q}_u$ was determined by trial and error. Similarly, N_p and N_c were determined by trial and error to obtain good control performance. In addition, adjusting the boom and arm valve openings to operate the boom and the arm was necessary. Thus, the relationship between each torque and valve opening was obtained through preliminary experiments. The torque command value was converted into the valve opening command value to operate the boom and arm using this relationship. When the bucket tip position passed the endpoint in the target trajectory, the excavation ended and u(t) = 0. Furthermore, an obstacle was embedded in advance at positions that would interfere with the excavation as shown in Fig. 10. It was a box covered with metal, approximately 10 times denser than the soil to be excavated. The obstacle was installed approximately 200 [mm] from the excavation start point and approximately 20 [mm] from the ground surface.

Figs. 11 and 12 show the results of method 1 and those of the proposed control system. The excavation operation progresses from left to right in the graph showing the bucket tip trajectory. Furthermore, a graph of the bucket tip trajectory is shown in Fig. 12, and the transition of the target trajectory during excavation is indicated by different colors. In method 1, as shown in Fig. 11, the bucket tip can excavate along the target trajectory up to the third point $r = 0.24$ [m] of the target trajectory without an obstacle. However, the excavation stopped at approximately $x_{tip}(t) = 0.67$ [m] when viewed from the excavation plane at approximately $y = 0.25$ [m] owing to the obstacle. In addition, the alarm was triggered before reaching the third point of the target trajectory and the system parameter $\hat{b}_0(t)$ became small. The hydraulic excavator-environment interaction model represents the ease with which excavation progresses, and small $b_0(t)$ indicates that the excavation is difficult to proceed. Since the excavation stopped after the alarm triggered, it can be seen that an obstacle made excavation difficult, and the interaction model captured this situation. On the other hand, in the proposed control system, as shown in Fig. 12, when viewed from the excavation plane, the bucket tip also moved along the preset target trajectory immediately after the start of the excavation, as shown in method 1. When the DD-MPC predicted that the excavation will stop around obstacles, the target trajectory was modified. This is indicated by the change of $r(t)$ in the graph of $y(t)$ in Fig. 12. For example, the trajectory was modified where $r(t)$ became small, such as around 1 [s] in the graph. In addition, the modified target trajectories are shown in green, cyan, magenta, and blue, and the changes in the target trajectory are shown by repeating this order in the bucket tip trajectory graph. Furthermore, the target trajectory was modified from smaller to larger $y_{tip}(t)$ coordinate. With this modification, excavation could be performed up to the endpoint of the eighth target trajectory without stopping the operation. In particular, the modification of the third and fourth target points near the obstacle significantly contributed to this result. Comparing the graphs of $y(t)$ for both methods, it can be observed that the control performance was improved by correcting the target trajectory. In this verification, it was confirmed that the excavation trajectory endpoint can be reached by modifying the target trajectory using prediction and decision of the interaction model. However, further research on the amount and number of modifications is necessary. For example, by considering the obstacle position in this experiment, which can be inferred from Fig. 11, the obstacle can be avoided without modifying a large upward movement, such as $x_{tip} = 0.66$ [m] in Fig. 12. Furthermore, because soil characteristics were limited, verification of its effectiveness under various conditions is necessary. Also, the proposed method has an issue with the tracking performance of the target trajectory. In particular, tracking performance was not good in the direction where y_{tip} became deeper at the excavation start, around $x = 0.75$ [m], and where the bucket tip was lifted at the excavation end, around $x = 0.35$–0.50 [m]. It was difficult to track the target trajectory due to the relationship between the hydraulic circuit characteristics of the RC excavator and the posture of the attachment required of tracking in the experiment. As a result, the bucket tip trajectories shown in Fig. 11 and Fig. 12 were obtained. On the other hand, it is necessary to verify the tracking performance when applying the proposed method to an actual hydraulic excavator. In some cases, the proposed method is needed to improve.

By applying the proposed control system to the combined motion of the boom and arm, it is possible to excavate along the target trajectory. It was also confirmed that the endpoint of the target trajectory could be reached by modifying the trajectory when an obstacle appears in the ground.

9. Conclusion

This study has proposed a control system for autonomous excavators to adapt to environmental conditions through decision by using data and models. Autonomous excavator operation is affected by environmental conditions, which change during work. Therefore, a control system that considers such changes is required. A hydraulic excavator-environment interaction model has been built, and DD-Modeling enabled the expression of environmental changes in the model. In addition, the DD-MPC enabled to calculate the control input and modification of the target excavation trajectory suitable for environmental conditions. Furthermore, by distributing the control input calculated by the DD-MPC, the proposed control system has been applicable to excavations by booms and arms. The control system has been verified using an RC excavator, which could adapt to environmental conditions and achieve a given excavation operation. On the other hand, the proposed method has some issues. One is the data reliability stored in database or used for modeling, because the modeling accuracy is affected by the data in database. In particular, persistent excitation is

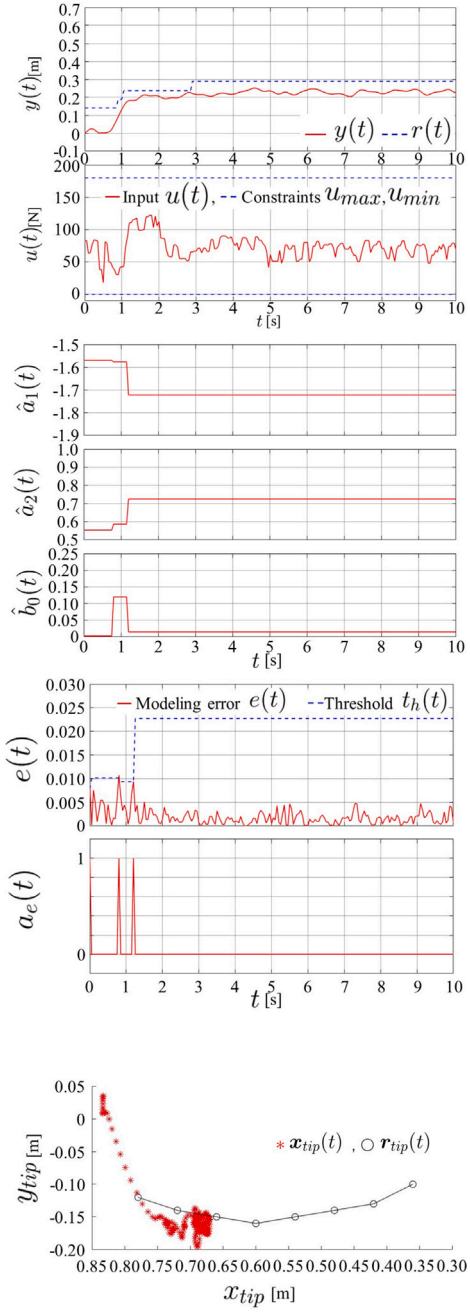


Fig. 11. Experimental result without trajectory planner (method 1).

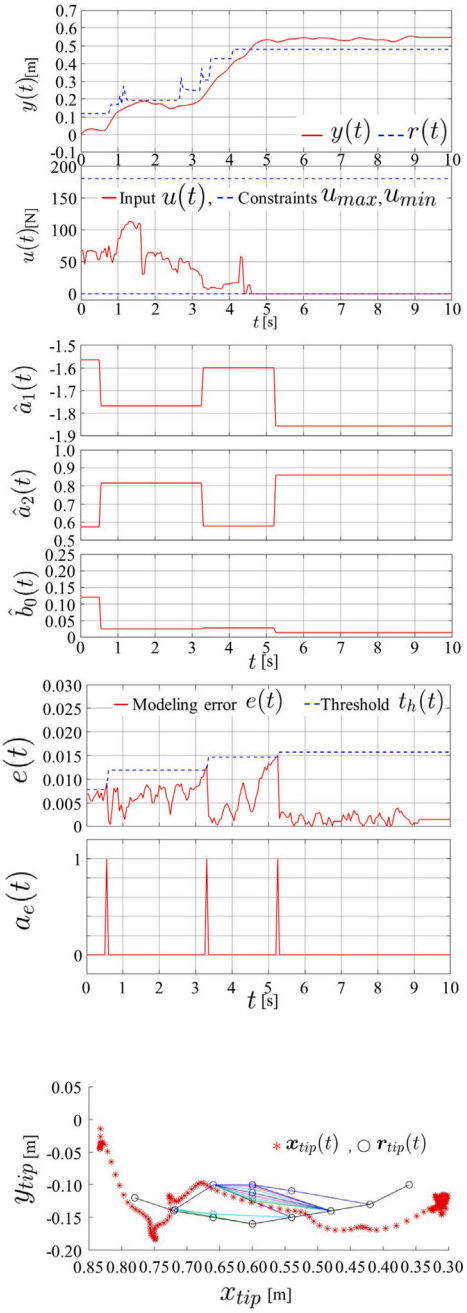


Fig. 12. Experimental result with trajectory planner.

important when identifying models to be stored in database, however, the proposed method has not taken it into account. The second is that it is needed the scheme to collect data from various construction sites and build models. The third is the calculation cost. High specification is required of the controller to be implemented since both MPC and DD-Modeling are incorporate.

The proposed control system will be extended to carry out the bucket operation in the future. Furthermore, the control system will be verified using actual excavators under various environmental conditions, such as at actual construction sites. In addition, further development of our research is expanded by extending the control system to cyber–physical system (CPS). CPS is a hot topic in control engineering in recent years. In Shin and Um (2023), a control system has been proposed and experimentally verified that includes three mechanisms to control a machine, to collect data, and to build models using data while operating the controlled object. Our proposed method needs these kinds of mechanisms to apply it to real hydraulic excavators that works in actual construction sites, and will be expanded to CPS.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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